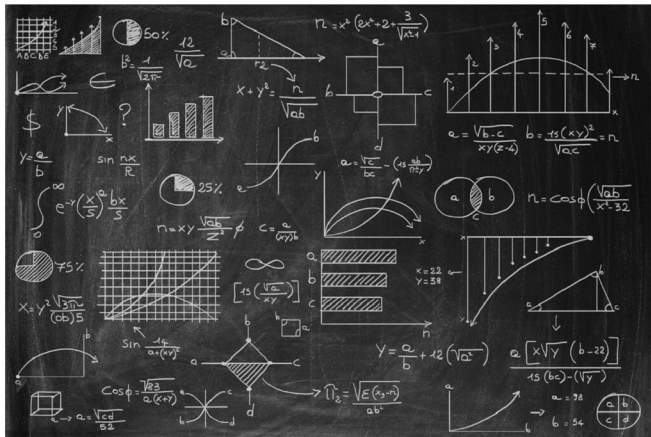


Maxima is a descendant of Macsyma, a breed of computer algebra systems, which was developed at MIT in the late 1960s. Owing to its open source nature, it has an active user community. This is the 17th article in the 'Mathematics in Open Source' series, in which the author deals with fundamental trigonometric expressions.



learning but rather to apply the formulae to get the various end results, assuming that you already know the formulae.

Maxima provides all the familiar fundamental trigonometric functions, including the hyperbolic ones (see Table 1).

Table 1

Mathematical names	Normal		Hyperbolic	
	Functions	Inv. Functions	Functions	Inv. Functions
sine (<i>sin</i>)	<i>sin()</i>	<i>asin()</i>	<i>sinh()</i>	<i>asinh()</i>
cosine (<i>cos</i>)	<i>cos()</i>	<i>acos()</i>	<i>cosh()</i>	<i>acosh()</i>
tangent (<i>tan</i>)	<i>tan()</i>	<i>atan()</i>	<i>tanh()</i>	<i>atanh()</i>
cosecant (<i>cosec</i>)	<i>csc()</i>	<i>acsc()</i>	<i>csch()</i>	<i>acsch()</i>
secant (<i>sec</i>)	<i>sec()</i>	<i>asec()</i>	<i>sech()</i>	<i>asech()</i>
cotangent (<i>cot</i>)	<i>cot()</i>	<i>acot()</i>	<i>coth()</i>	<i>acoth()</i>